

Fourth-Order Radiation of Pure-Weyl Black Holes

Exact Branch Splitting, Horizon Flux, and a Causal Truncation No-Go

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Abstract

We test whether the Einstein radiative branch of four-dimensional pure Weyl gravity can be selected by ordinary causal boundary conditions around a black hole. We first classify an exact static spherical Bach-flat family within a declared Laurent ansatz, reproduce the Mannheim–Kazanas solution, and derive a residual-basic horizon generator. On the static parameter slice its Lee–Wald Hamiltonian, Wald entropy, and temperature satisfy an exact first law at every simple horizon. Arbitrary time-dependent spherical Weyl and diffeomorphism perturbations are null charge directions at linear order.

We then study Schwarzschild perturbations. In the axial $\ell = 2$ sector the fourth-order Bach equation factors exactly through the linearized Ricci tensor:

$$\delta B_{ab} = \frac{1}{2} \square(\delta R_{ab}) + C_{acbd}(\delta R)^{cd}.$$

The Einstein/Regge–Wheeler branch is $\delta R_{ab} = 0$; the extra branch is a second-order Lichnerowicz-type wave for $\psi_{ab} = \delta R_{ab}$. The extra branch has a two-parameter analytic ingoing family at the future horizon, is luminal and bounded at infinity for real frequency, and has the same leading asymptotic characteristics as the Einstein branch. The action-derived Lee–Wald current makes the Einstein self-block null for conjugate waves, while independently verified fixtures give nonzero Einstein–extra and extra–extra horizon flux.

It follows that horizon regularity plus local causal decay cannot truncate the axial fourth-order exterior to its Einstein branch. Such a truncation would require an additional projection on scattering data rather than a local initial-boundary condition. In the polar $\ell = 2$ sector we prove the corresponding exact trace-coupled branch split; its horizon reach, flux, and causal disposition remain the next theorem gate. We make no claim about complex frequencies, stability, quasinormal spectra, nonlinear evolution, Hawking radiation, or quantum unitarity.

AI authorship and computational accountability

The named AI systems developed the programme, derivations, symbolic implementations, independent verification architecture, claim audits, and this manuscript. Asger Alstrup Palm commissioned and coordinated the work and is the corresponding human contact. Every certified statement below is bound to a machine-readable exact certificate and a separately implemented verifier. The cross-flux result uses controlled high-precision series evaluation with exact recurrence data and independent frequencies; all local tensor identities and polynomial classifications are exact. The manuscript remains conditional on final human review.

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1 The physical question

Fourth-order gravitational equations contain more local solutions than the Einstein equations. A common proposal is to retain the Einstein branch by a boundary condition. Around a Lorentzian black hole, that proposal must pass a causal test:

Can regular initial data and local conditions at the future horizon and outer boundary remove the extra branch while retaining ordinary Einstein gravitational waves?

The answer in the certified Schwarzschild axial $\ell = 2$ laboratory is no. The extra branch is regular at the future horizon, propagates on the same real characteristics as the Einstein branch, has no growing real- frequency asymptotic solution, and carries nonzero action-derived horizon flux. Ordinary local regularity and decay therefore admit it.

This conclusion is narrower than a statement about the full theory. It is an axial, linear, real-frequency causal-disposition theorem. The polar branch has been split but not yet propagated through the same chain, and no complex-frequency stability analysis has been performed.

1.1 Claims and dependency tags

The static family and local operator identities carry LOCAL-ALGEBRAIC. Harmonic branch and current statements also carry REDUCED-MODE. The final axial disposition uses the declared horizon

and asymptotic domains and is a scoped LORENTZIAN-CAUSAL no-go. It is not a complete exterior Green-hyperbolicity theorem.

2 Pure Weyl gravity and its covariant current

We use signature $(-+++)$ and action

$$S_W[g] = \alpha \int_M \sqrt{-g} C_{abcd} C^{abcd} d^4x. \quad (1)$$

The field equation is the Bach equation $B_{ab} = 0$, with convention

$$B_{ab} = \nabla^c \nabla^d C_{acbd} + \frac{1}{2} R^{cd} C_{acbd}. \quad (2)$$

The covariant presymplectic current is obtained from the literal variation of the curvature-momentum potential of this action. Its normalization is fixed by the off-shell Green identity

$$\partial_t F^t(h_A, h_B) + \partial_r F^r(h_A, h_B) = 4\alpha \int_{S^2} \sqrt{q} (h_B^{ab} \delta B_{ab}[h_A] - h_A^{ab} \delta B_{ab}[h_B]). \quad (3)$$

Thus the current is conserved for pairs of linearized Bach solutions and is not a freely rescaled master-equation Wronskian.

3 Static spherical Bach-flat family

In Schwarzschild gauge,

$$ds^2 = -B(r)dt^2 + B(r)^{-1}dr^2 + r^2 d\Omega_2^2, \quad (4)$$

the declared Laurent ansatz is

$$B(r) = w - \frac{u}{r} + \gamma r - kr^2 + \frac{c_2}{r^2} + c_3 r^3. \quad (5)$$

Theorem 3.1 (Complete Bach-flat locus in the Laurent ansatz). *The metric is Bach-flat if and only if*

$$c_2 = c_3 = 0, \quad w^2 + 3u\gamma = 1. \quad (6)$$

The Mannheim–Kazanas parametrization is recovered by

$$w = 1 - 3\beta\gamma, \quad qquad u = \beta(2 - 3\beta\gamma).$$

Within this family the metric is Einstein if and only if $\gamma = 0$.

Completeness here refers only to the Laurent ansatz. The certificate does not claim a new proof of the global static spherical classification.

The form-preserving local $\text{Diff} \times \text{Weyl}$ transformations act as dilation and translation on the cubic

$$Q(x) = -ux^3 + wx^2 + \gamma x - k, \quad qquad dx = r^{-1}.$$

Their orbit rank is two. A single continuous invariant may be taken as

$$J = u^2 \text{disc}(Q),$$

while $\text{disc}(Q) = 0$ is the degenerate-horizon locus. The conformal translation may fail globally where its Weyl factor vanishes; it is used only as a local orbit statement.

An exact non-Einstein three-horizon fixture is

$$(\beta, \gamma, k) = \left(\frac{3}{2}, \frac{12}{19}, \frac{1}{19} \right), \quad rB(r) = -\frac{1}{19}(r-1)(r-3)(r-8).$$

All curvature invariants are finite at the simple horizons. In ingoing Eddington–Finkelstein coordinates,

$$ds^2 = -B(r)dv^2 + 2dvdr + r^2 d\Omega_2^2,$$

the metric and Bach equation are regular there.

4 Residual-basic energy and the static first law

For the chart generator ∂_t , the bare static Lee–Wald one-form is not integrable. The obstruction disappears when the generator is normalized consistently with the residual quotient.

Theorem 4.1 (Normalized static generator). *Let*

$$\chi = u \partial_t, \quad u = \beta(2 - 3\beta\gamma). \quad (7)$$

Up to a function of the single residual invariant J , this normalization is forced by residual basicness and dilation weight. Its Hamiltonian is

$$H = -16\pi\alpha\beta^2 D_2, \quad D_2 = 9\beta^2\gamma^2 k - \beta\gamma^3 - 12\beta\gamma k + \gamma^2 + 4k. \quad (8)$$

At every simple horizon r_h , the Wald entropy

$$S_h = 64\pi^2\alpha\beta \frac{2 - 3\beta\gamma + \gamma r_h}{r_h} \quad (9)$$

and normalized temperature

$$T_h = \frac{uB'(r_h)}{4\pi} \quad (10)$$

satisfy

$$dH = T_h dS_h \quad (11)$$

identically modulo $B(r_h) = 0$.

This is a theorem on the static parameter slice, not yet a general physical-process first law. Its linear dynamical extension nevertheless passes a strong gauge audit: arbitrary time-dependent spherical Weyl rescalings and diffeomorphisms have exactly zero charge and flux, the entropy is conformally invariant on the symbolic family, and the normalized generator extension is unique at linear order under the declared boundary scalar rule.

5 Axial Schwarzschild branch splitting

We now specialize the background to Schwarzschild and use axial Regge–Wheeler gauge,

$$h_{t\phi} = h_0(t, r)S_{\ell m}, \quad h_{r\phi} = h_1(t, r)S_{\ell m}.$$

The $\ell = 2$ linearized Bach tensor is fourth order, trace-free, and obeys the exact linearized Noether identity. Eliminating the constraint on the Einstein subbranch reproduces the Regge–Wheeler equation for $\Psi = Bh_1/r$,

$$B(B\Psi)' + (\omega^2 - V_{\text{RW}})\Psi = 0, \quad V_{\text{RW}} = B \left(\frac{6}{r^2} - \frac{6m}{r^3} \right). \quad (12)$$

Theorem 5.1 (Axial composition identity). *On a Ricci-flat background, for the certified axial carrier,*

$$\boxed{\delta B_{ab} = \frac{1}{2}\square(\delta R_{ab}) + C_{acbd}(\delta R)^{cd}.} \quad (13)$$

Consequently the Einstein branch $\delta R_{ab} = 0$ injects into the Bach kernel, while the extra branch is the second-order system

$$\frac{1}{2}\square\psi_{ab} + C_{acbd}\psi^{cd} = 0, \quad \psi_{ab} := \delta R_{ab}. \quad (14)$$

The identity is componentwise and off shell in the perturbation. It does not extend in this two-term form to the certified non-Einstein static fixture. Thus the clean branch split is a Ricci-flat-background theorem.

6 The extra branch reaches the future horizon

In ingoing coordinates (v, r) , Fourier modes are written $e^{i\omega v}$. The extra carrier’s divergence constraint solves its third component polynomially; the apparent Schwarzschild-coordinate pole is a chart artifact.

Theorem 6.1 (Analytic ingoing family). *For axial $\ell = 2$ and every frequency ω , the future horizon $r = 2m$ is a regular singular point of the extra first-order radial system. Its residue spectrum is*

$$\{0, 0, -4im\omega, -2 - 4im\omega\}, \quad (15)$$

and the zero eigenspace has dimension two. Hence there is a two-parameter family of extra-branch solutions analytic at the future horizon, with no leading logarithm.

Future-horizon regularity therefore does not select the Einstein branch.

7 Action-derived horizon flux

Substituting two on-shell Regge–Wheeler solutions into the current (3) gives

$$F_{EE}^r = -\frac{192\pi\alpha(\omega_1^2 - \omega_2^2)}{5\omega_1\omega_2 r}\Psi_1\Psi_2. \quad (16)$$

For conjugate pairs $\omega_2 = \pm\omega_1$, this vanishes identically.

Theorem 7.1 (Axial flux disposition). *On Schwarzschild, axial $\ell = 2$:*

1. *the Einstein/Regge–Wheeler self-block is symplectically null for conjugate wave pairs;*
2. *ingoing extra modes have a nonzero extra–extra horizon-flux pairing;*
3. *the Einstein–extra cross pairing is nonzero.*

The last two statements are certified at $m = 1$ and independent real frequencies $\omega = 3/5, 2/7, 1/2$, using order-16 ingoing series and radius-stability checks. The exact Regge–Wheeler null result is the in-run control.

For $\alpha > 0$, the certified ingoing extra fixtures obey $iF_{XX}^r > 0$ in the declared convention. We do not promote that classical sign to a Hilbert norm. The result needed here is nondegeneracy: the extra branch is not a zero-flux artifact.

8 Asymptotic propagation and causal disposition

At real frequency and large radius, the extra carrier has dispersion

$$(\lambda_r^2 - \omega^2)^2 = 0 \quad (17)$$

and growth exponents

$$\sigma \in \{\pm 2im\omega, \pm 2im\omega - 1\}. \quad (18)$$

Thus it is luminal, with Coulomb logarithmic phases and amplitudes of order r^0 or r^{-1} . There is no growing real-frequency asymptotic solution. The Einstein control has the same leading characteristics and logarithmic phases.

Theorem 8.1 (Axial causal truncation no-go). *On the declared Schwarzschild axial $\ell = 2$, real-frequency mode domain, no boundary prescription based only on future-horizon regularity and local causal decay at the outer boundary excludes the extra branch while retaining the Einstein branch. Selecting the Einstein branch requires an additional branch projection on scattering data, rather than a local causal initial-boundary condition.*

Proof. The extra branch has a two-dimensional ingoing-analytic horizon family, so horizon regularity admits it. Its real-frequency outer solutions are bounded and have the same characteristic falloff class as the Einstein solutions, so local asymptotic decay does not distinguish it. Its self and mixed Lee–Wald fluxes are nonzero, so it cannot be discarded as a radical direction. A condition that removes its independent scattering amplitudes must therefore project branch data beyond these local endpoint conditions. $\square$

The conclusion is compatible with Euclidean or two-boundary Einstein-branch selection: such a prescription may define a sector. The theorem says that the sector is not generated by ordinary one-ended Lorentzian regularity and decay in this axial laboratory.

9 Polar branch: exact split and open causal chain

For polar $\ell = 2$ Regge–Wheeler-gauge perturbations, the trace variation δR need not vanish. The axial formula acquires two universal trace terms.

Theorem 9.1 (Polar trace-coupled composition identity). *On a Ricci-flat background,*

$$\delta B_{ab} = \frac{1}{2}\square(\delta R_{ab}) + C_{abcd}(\delta R)^{cd} - \frac{1}{6}\nabla_a\nabla_b\delta R - \frac{1}{12}g_{ab}\square\delta R. \quad (19)$$

The Einstein polar branch injects by $\delta R_{ab} = 0$, while the extra branch is a second-order trace-coupled Lichnerowicz system for $(\delta R_{ab}, \delta R)$.

This opens, but does not complete, the polar analogue of the axial chain. The Zerilli benchmark, horizon indicial system, action-derived flux blocks, outer asymptotics, and causal disposition remain pending. No parity-complete black-hole truncation theorem is claimed until those steps close.

10 Result ledger

Question	Certified answer	Scope
Static Bach-flat black holes?	exact Laurent-family classification and three-horizon fixture	local/static
Differentiable energy?	residual-basic normalized generator, exact static first law	static slice; linear spherical gauge audit
Einstein axial waves present?	yes; exact Regge–Wheeler injection	Schwarzschild, axial $\ell = 2$
Extra axial waves present?	yes; second-order Ricci carrier	same
Reach future horizon?	yes; two analytic ingoing amplitudes	mode level
Carry current?	nonzero mixed and extra flux; Einstein self-block null	exact identity plus three frequency fixtures
Removed by local causal boundaries?	no	real-frequency axial domain
Polar split?	exact trace-coupled identity	causal chain still open
Stable/ringing?	not established	complex frequencies open

11 What is not proved

The paper does not establish:

- a complete exterior initial-boundary well-posedness theorem;
- general ℓ, m , the completed polar chain, or non-Einstein background branch splitting;
- complex-frequency poles, mode stability, quasinormal modes, or ringdown waveforms;
- a second-order physical-process first law or nonlinear horizon dynamics;
- asymptotically flat conformal-gravity scattering with a complete boundary phase space;
- Hawking states, particles, a quantum BRST quotient, positivity, or unitarity.

In particular, “causally unavoidable” means unavoidable by the declared local axial boundary tests, not that every nonlinear or quantum state must contain the branch.

12 Certificate and reproduction ledger

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13 Outlook

The next calculation is not a generic stability claim. It is the polar analogue of the exact axial sequence:

Zerilli control $\longrightarrow$ horizon reach $\longrightarrow$ Lee–Wald flux $\longrightarrow$ outer disposition.

Only then should complex-frequency structure and quasinormal boundary conditions be activated. A later asymptotic phase-space construction must decide which generator is physical, which flux reaches null infinity, and whether the extra branch survives a globally differentiable charge algebra.

The present result already fixes an important baseline:

The extra axial branch is regular, radiative, and not removable by ordinary local causal boundary conditions.

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